



# Exploratory Data Analysis

Assessment 1 · Group030

FIT5196 Data Wrangling

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# 1. Data and analytical approach

## 1.1 Start from the submitted relational data

The analysis begins by loading the six standardised CSV files produced in the solution notebook. The tables contain 5,000 orders, 15,723 order-item lines, 500 customers, 5,000 deliveries, 1,000 products and 7,000 reviews. Identifiers remain strings; only the numeric, date and Boolean fields used below are converted after loading.

| Table           | Rows   | Columns | Role in the EDA                  |
|-----------------|--------|---------|----------------------------------|
| orders          | 5,000  | 23      | Order value, discount and time   |
| order_items     | 15,723 | 6       | Product-level sales contribution |
| customers       | 500    | 20      | Prior customer activity          |
| deliveries      | 5,000  | 20      | OTIF and fulfilment              |
| products        | 1,000  | 21      | Category and catalogue cost      |
| product_reviews | 7,000  | 21      | Rating and review timing         |

## 1.2 Check the data before analysis

Before choosing the figures, we rechecked that the row count of each table matched its primary-key count. All six checks passed. For relational figures, the parent key was required to be unique and the left-hand row count had to remain unchanged. Orders were aggregated before the customer analysis, and multiple reviews were averaged within order before the delivery join. These checks protect the observation unit used in each chart.

The solution notebook provides the complete transformation and validation record. For this EDA, the most relevant results are zero orphan keys across the submitted relationships, a \$0.00 maximum arithmetic difference, consistent delivery and OTIF fields, and preservation of the 303 reviews containing non-Latin script.

## 1.3 Move from table structure to analytical questions

We first examined order-value distribution and composition, then customer groups, product categories and time. We next tested review behaviour, multilingual text measurement, delivery operations and the delivery-review relationship. This sequence progresses from single-table description to checked relational analysis.

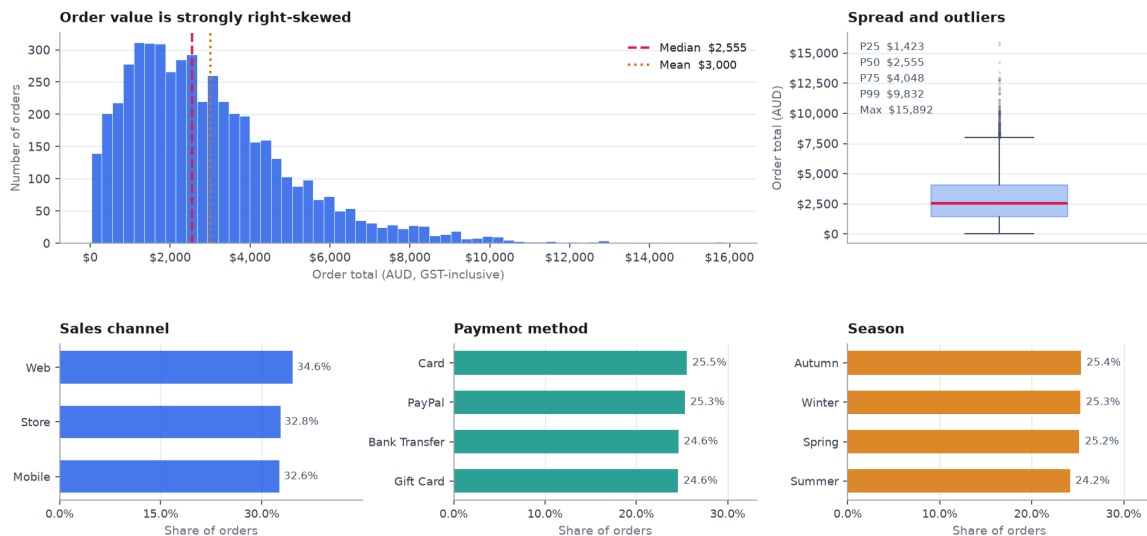
Continuous group means are reported with 95% confidence intervals, and OTIF proportions use Wilson 95% intervals. The figures describe associations in this export and are not interpreted as causal effects.

## 1.4 Assessed EDA coverage

| Requirement                         | Evidence              | Observation unit                 |
|-------------------------------------|-----------------------|----------------------------------|
| Univariate distribution/composition | Figure 1              | Order                            |
| Bivariate comparison                | Figures 2 and 5       | Order, customer or review        |
| Multivariate/segmented analysis     | Figures 3, 7 and 8    | Item, delivery or review         |
| Temporal pattern                    | Figure 4              | Order aggregated by time         |
| Review/text behaviour               | Figures 5, 6 and 8    | Review                           |
| Delivery/operational performance    | Figures 7 and 8       | Delivery or review               |
| Checked relational analysis         | Figures 2, 3, 7 and 8 | Order, item, delivery and review |

## 2. Assessed visualisations

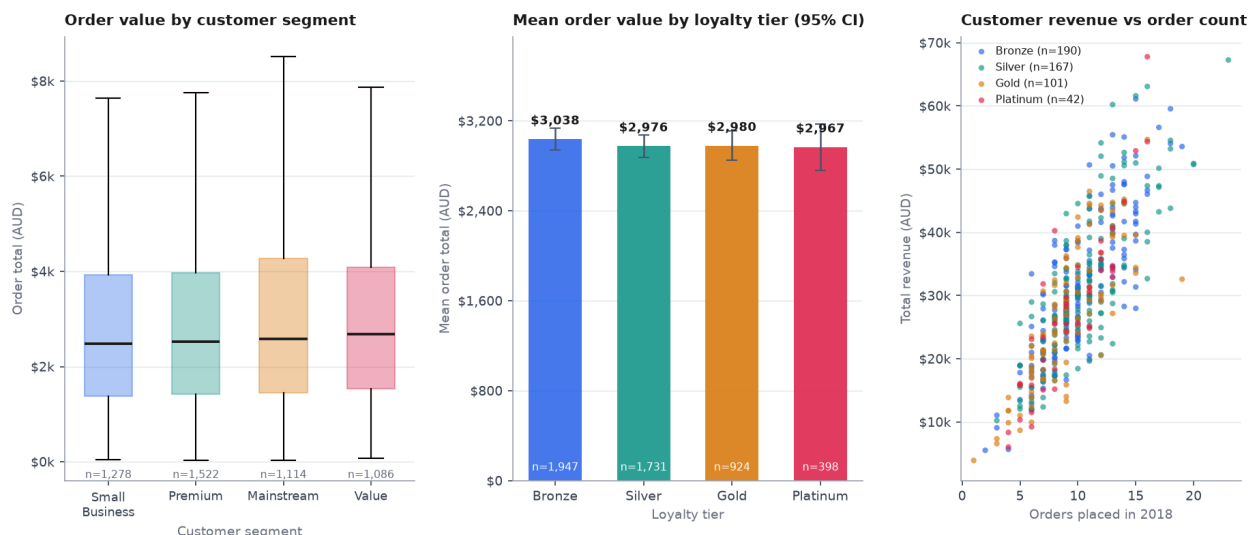
### 2.1 How are order values distributed, and how is order volume composed?



**Observation unit and denominator.** One canonical order; 5,000 orders. **Tables and join keys.** orders; no join. **Why this visualisation.** A histogram and boxplot retain the skew and outliers, while aligned percentage bars compare three compositions using a common order denominator. **Interpretation and explanation.** The median is more representative than the mean because a small high-value tail pulls the mean upward. Near-uniform channel, payment and season shares show that order counts are not concentrated in one route. **Material limitation.** Order totals combine baskets containing different numbers of items, and equal order shares do not imply equal revenue shares.

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### 2.2 Do customer segment and loyalty tier explain order value?



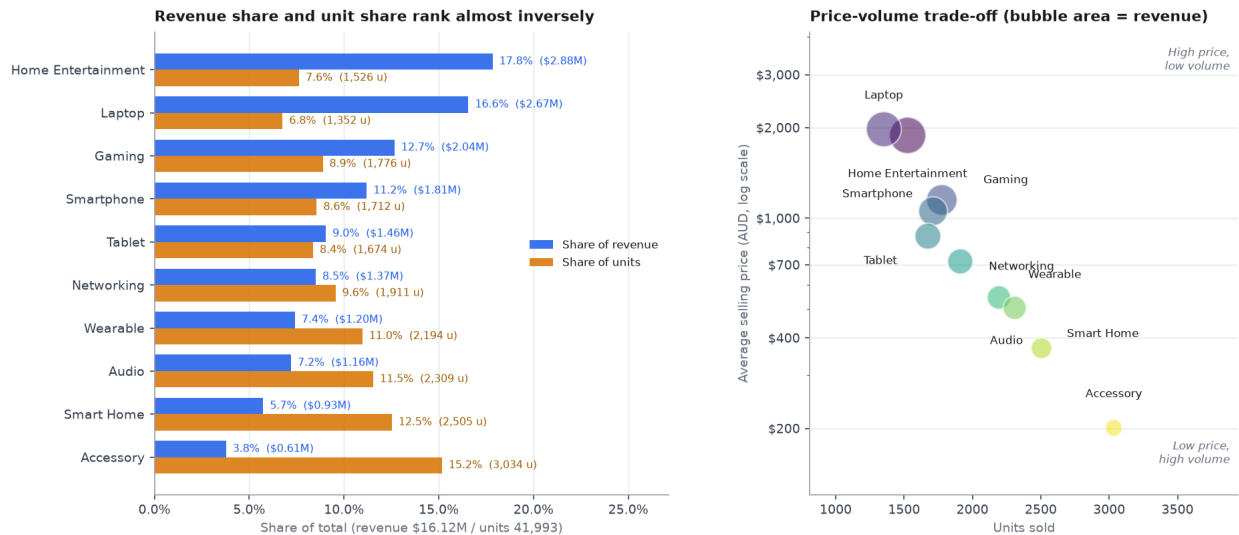
**Observation unit and denominator.** Left and centre: one order, 5,000 orders; right: one customer, 500 customers. **Tables and join keys.** orders joined to customers on customer\_id (many-to-one); the order count must remain 5,000. **Why this visualisation.** Boxplots show the full within-segment spread, mean points/bars with 95% intervals show uncertainty by tier, and a customer-level scatter separates cumulative revenue from order frequency.

**Interpretation and explanation.** Segment medians and loyalty-tier means are close relative to within-group variation. Cumulative customer revenue rises with the number of recorded orders, but this is descriptive and partly mechanical because revenue accumulates across orders. **Material limitation.** Repeated orders from the same customer are not independent, and tier is observed at export rather than at each order date.

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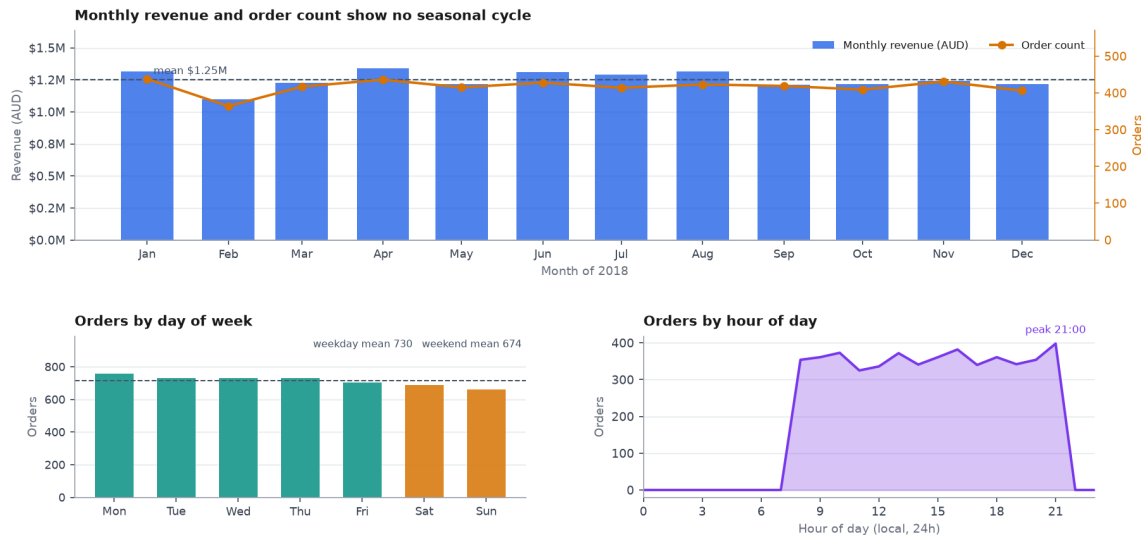
## 2.3 Which categories drive revenue, and is that volume or price?



**Observation unit and denominator.** One order-item line; 15,723 lines, 41,993 units and \$16.12 million line revenue. **Tables and join keys.** `order_items` joined to `products` on `product_id` (many-to-one); row count and revenue are checked after joining. **Why this visualisation.** Paired horizontal bars compare revenue and unit shares on the same scale. The price-volume bubble chart adds average selling price and revenue without summing order-level fields at item grain. **Interpretation and explanation.** High-priced categories generate large revenue shares from relatively few units, whereas accessories have high volume but low revenue share. Price point therefore explains much of the category ranking in this period. **Material limitation.** Revenue is not profit; cost, returns and promotions could change the commercial ranking.

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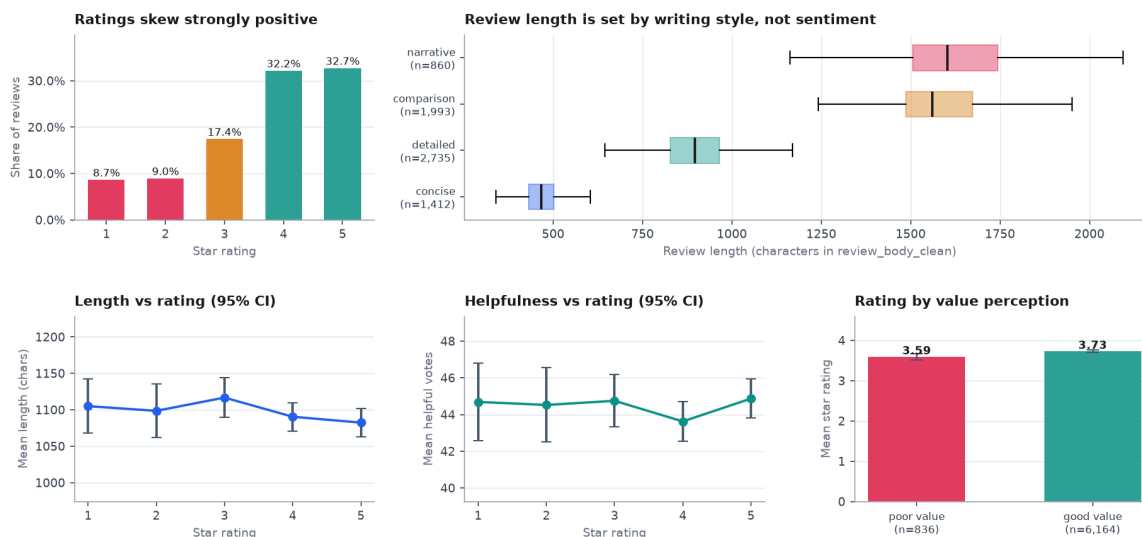
## 2.4 How does demand move through the year, the week and the day?



**Observation unit and denominator.** One order aggregated by month, weekday or hour; 5,000 orders from 1 January to 31 December 2018. **Tables and join keys.** `orders`; no join. **Why this visualisation.** Coordinated time panels distinguish annual, weekly and intraday structure. Bars make period totals readable and the hourly area-line plot shows concentration through the day. **Interpretation and explanation.** Month and weekday volumes vary little, while orders concentrate in trading hours and fall sharply overnight. Intraday timing is therefore more relevant to short-term staffing than a seasonal narrative. **Material limitation.** One year cannot separate stable seasonality from a one-off pattern, and timestamps have no timezone.

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## 2.5 What drives review length, helpfulness and star rating?



**Observation unit and denominator.** One canonical product review; 7,000 reviews. **Tables and join keys.** `product_reviews`; no join. **Why this visualisation.** A rating composition chart, writing-style boxplots and mean comparisons with 95% intervals distinguish distribution, text length and labelled experience. **Interpretation and explanation.** Ratings are strongly positive. Review length differs sharply by supplied writing style but changes little across ratings, and helpful votes are similarly flat. The supplied value-experience label has only a modest rating difference. **Material limitation.** Writing style and value experience are source labels, not independently inferred text features, while helpful votes lack a views/exposure denominator.

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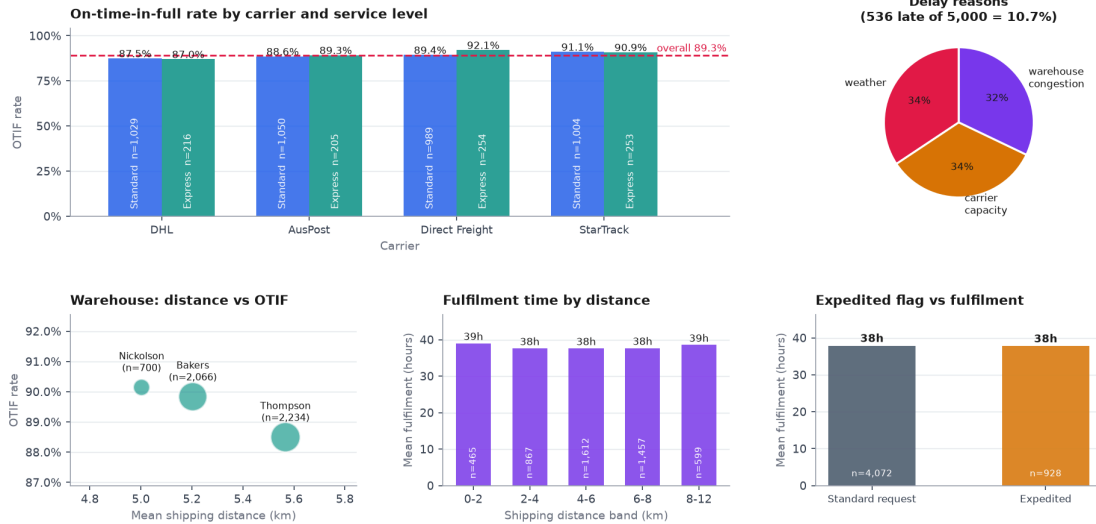
## 2.6 Does script class distort the review-length measures?



**Observation unit and denominator.** One review; 7,000 reviews, including 303 containing non-Latin script. **Tables and join keys.** `product_reviews`; no join. **Why this visualisation.** A log-scale language count handles the large English/non-English imbalance; overlaid distributions and paired means compare character and whitespace-token measures. **Interpretation and explanation.** Non-Latin reviews have a different relationship between characters and whitespace tokens. Character length is therefore the safer cross-script descriptive measure, while whitespace word counts should be interpreted cautiously. **Material limitation.** A binary script flag combines languages with different writing systems and tokenisation rules; 303 reviews provide limited subgroup precision.

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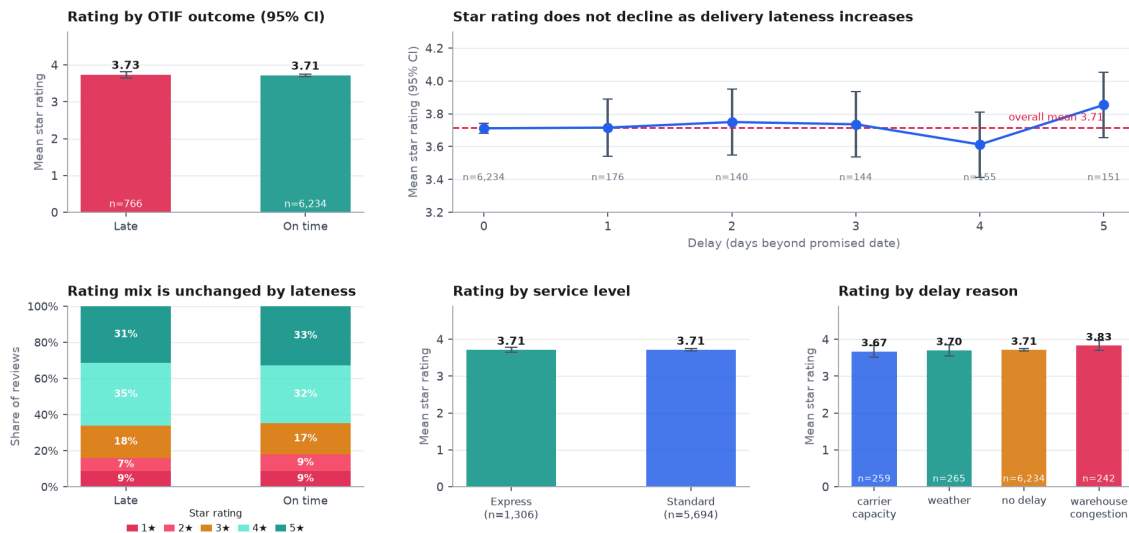
## 2.7 Where does delivery performance actually vary?



**Observation unit and denominator.** One completed delivery; 5,000 deliveries, of which 536 (10.7%) are late. **Tables and join keys.** deliveries joined to orders on order\_id (one-to-one); row count remains 5,000. **Why this visualisation.** Grouped bars compare carrier/service OTIF rates, while coordinated panels test delay reasons, warehouse distance, fulfilment by distance and expedited requests. **Interpretation and explanation.** OTIF varies by only a few percentage points across carriers and service levels; delay reasons are near-even and fulfilment averages remain close across the observed distance bands. There is no single dominant operational split in this export. **Material limitation.** Distances cover only a narrow metropolitan range and carrier assignment is observational, so small gaps cannot be interpreted as carrier effects.

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## 2.8 Does delivery performance shape customer ratings?



**Observation unit and denominator.** One review; 7,000 reviews covering 4,044 distinct orders. **Tables and join keys.** product\_reviews joined to deliveries on order\_id (many-to-one); the review count remains 7,000. **Why this visualisation.** Rating means with 95% intervals, full rating composition and operational subgroups test whether a small mean difference hides a distributional change. **Interpretation and explanation.** Rating means and the 1–5 star mix are almost unchanged by OTIF, delay days, service level and delay reason. The result supports a weak association in this observed reviewer sample, not evidence that delivery quality never matters. **Material limitation.** Reviews from the same order are not independent and only reviewed orders are observed; the displayed review-level confidence intervals may therefore be optimistic.

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### 3. Evidence-based findings

- 3.1 A typical order is materially below the mean.** At order grain (5,000 orders), Figure 1 shows a median total of **\$2,555** versus a mean of **\$3,000**, with the 99th percentile near **\$11,823**. The right tail can overstate normal basket value if the mean is used alone. Different line-item counts are an alternative explanation for part of the tail, so operational reporting should pair median and upper percentiles rather than treating the mean as a customer norm.
- 3.2 No sales channel dominates order count.** Figure 1 shows each of Cart, Store and Mobile contributing approximately **32–35% of 5,000 orders**. This reduces immediate dependence on one channel by volume, but the figure does not compare revenue, margin or acquisition cost. Channel strategy should therefore be evaluated with value and cost measures before reallocating investment.
- 3.3 Customer labels provide little separation in basket value.** Across 5,000 orders joined many-to-one to 500 customers, Figure 2 shows customer-segment medians within about **\$203** and loyalty-tier means around **\$2,947–\$3,038**, with overlapping 95% intervals. Within-group variation and repeated customers create uncertainty, so these labels should not be used alone for basket-value targeting; order context and product mix merit testing.
- 3.4 Current-period customer revenue mainly tracks recorded order frequency.** At customer grain (500 customers), Figure 2 shows cumulative revenue rising closely with the number of 2018 orders and little visible loyalty-tier separation. This is commercially relevant for retention planning, but the relationship is partly mechanical because revenue is summed across orders and cannot establish that frequency causes value. Future modelling should predict future frequency using only information known before the prediction window.
- 3.5 Category revenue and unit volume rank differently.** At order-item grain (15,723 lines; 41,993 units; **\$16.12m** revenue), Figure 3 shows Home Entertainment and Laptop producing about **34.1% of revenue from 14.4% of units**, while Accessory has the largest unit share but only about **\$0.61m** revenue. Price point explains much of this contrast, but revenue excludes cost and returns. Stock decisions should jointly consider units, revenue and contribution margin.
- 3.6 Joining at the wrong grain would materially inflate an order metric.** The Figure 3 control shows that summing `order_total` after the orders-items fan-out would produce approximately **\$53.19m**, or **3.55 times** the correct **\$15.00m** order total. This is a relational calculation risk rather than a market pattern. Order metrics must be calculated before item-level joins and reconciled to a control total.
- 3.7 Demand structure is intraday rather than strongly seasonal in this extract.** Across 5,000 orders in 2018, Figure 4 shows monthly counts between **406 and 438** (a **7.9%** peak-to-trough band) and similar weekday/weekend rates, while order counts collapse overnight. One year and unknown timezone limit generalisation. Staffing experiments should begin with hour-of-day demand and be revalidated on later periods.
- 3.8 Review ratings are positive, but text length is mainly a writing-style property.** Among 7,000 reviews, Figure 5 shows **64.9%** at 4–5 stars versus **17.6%** at 1–2 stars; median length is about **433 characters** for concise and **1,572** for narrative reviews, while mean length and helpful votes are broadly flat across ratings. Because style is supplied rather than inferred and helpful votes lack exposure, text models should preserve style controls and avoid treating raw vote totals as intrinsic quality.
- 3.9 Whitespace word counts are not directly comparable across scripts.** Figure 6 covers 7,000 reviews, including **303 non-Latin-script reviews**, and shows that character length and whitespace-token count move differently by script class. The small and heterogeneous subgroup makes the exact difference uncertain, but it is large enough to affect feature engineering. Character-based or language-aware text measures should be preferred, with performance reported by language/script group.
- 3.10 Observed delivery variation does not translate into a clear rating difference.** At delivery grain, Figure 7 reports **89.3% OTIF across 5,000 deliveries** and only a **3.7 percentage-point** carrier spread; at review grain, Figure 8 shows late and on-time ratings of about **3.73 and 3.71** across 7,000 reviews with overlapping intervals. Carrier mix, clustered reviews and review-selection bias remain plausible explanations. Operational improvement should target service outcomes directly rather than assuming this cross-sectional rating signal measures customer impact.

### 4. Future machine-learning questions

No model is trained in this assignment. Each question is grounded in an EDA result and uses predictors available at the proposed decision time.

#### 4.1 OTIF-failure classification

**Use case, unit and target.** Before dispatch, rank **orders** for intervention; target `on_time_in_full = False`.

**Decision-time predictors.** Warehouse, carrier, service level, sales channel, expedited flag, order timestamp, item count/value, product weight and destination fields available before dispatch.

**Validation and metrics.** Forward time split by order date; PR-AUC and recall at a fixed intervention capacity, with ROC-AUC secondary.

**Risk and EDA grounding.** Exclude delivered date, delay days/reason and realised fulfilment fields. Carrier assignment may encode operational policy. Figure 7 grounds the 10.7% event rate and weak simple group separation.

## 4.2 Fulfilment-hours regression

**Use case, unit and target.** At order acceptance, estimate **hours per order** to support workload promises; target `fulfilment_hours`.

**Decision-time predictors.** Order time, warehouse, channel, service level, expedited flag, item count, basket value, product mix, total weight and destination features available at acceptance.

**Validation and metrics.** Rolling/forward time split; MAE plus 90th-percentile absolute error.

**Risk and EDA grounding.** Do not use dispatch/delivery dates or post-dispatch delay fields. Process changes can cause temporal drift. Figure 7 shows that distance bands and expedited status alone explain little variation.

## 4.3 Low-rating classification

**Use case, unit and target.** At review invitation time, identify **reviewed orders** at risk of a 1–2 star rating; target whether order-level mean rating is at most 2.

**Decision-time predictors.** Pre-invitation order, product/category, customer-history and completed-delivery fields; text and the eventual rating are excluded.

**Validation and metrics.** Forward split by delivery/review date, grouped by customer; PR-AUC, Brier score and recall at fixed contact capacity.

**Risk and EDA grounding.** Review propensity creates selection bias; sensitive customer proxies require subgroup checks. Figures 5 and 8 show class imbalance and weak delivery-only separation.

## 4.4 Future customer-frequency regression

**Use case, unit and target.** At a period boundary, forecast **next-period order count per customer** for retention planning; target future count.

**Decision-time predictors.** Only customer attributes and aggregates from periods ending before the boundary: recency, historical frequency/value, channel mix, category breadth and prior delivery experience.

**Validation and metrics.** Rolling-origin temporal validation; MAE and Poisson deviance, compared with a recency-frequency baseline.

**Risk and EDA grounding.** Random splits and lifetime fields can leak the future; new customers need a fallback. Figure 2 shows current revenue is closely associated with frequency but tier alone adds little visible separation.

## 4.5 Product clustering

**Use case, unit and target.** Group **products** for assortment and inventory review; unsupervised objective: stable, interpretable product segments.

**Decision-time predictors.** Catalogue attributes plus pre-cut-off price, cost, weight and historical unit/revenue summaries; scale numeric features and encode category/brand carefully.

**Validation and metrics.** Fit on an earlier period and assess stability on a later period using adjusted Rand index; silhouette score and business interpretability as secondary criteria.

**Risk and EDA grounding.** Dominant category labels and scaling can predetermine clusters; rare products may be unstable. Figure 3 motivates separating high-price/low-volume from low-price/high-volume products.

# 5. Limitations and conclusion

The analysis describes one 2018 export and does not establish causal effects. Several intervals treat rows as independent even though customers can place several orders and an order can have several reviews; the related figures state this limitation explicitly. Review-based results also apply only to orders that generated reviews. Despite these constraints, the eight figures cover all six required EDA categories, use all six tables, and include four checked relational analyses. The main decision value lies in respecting table grain, separating price from volume, focusing operational planning on intraday demand, and avoiding strong targeting claims from weak group differences.



## References

National Institute of Standards and Technology. (n.d.). *Histogram*. In *NIST/SEMATECH e-Handbook of statistical methods*. <https://www.itl.nist.gov/div898/handbook/eda/section3/histogra.htm>

External material informed general chart selection only. All numerical claims, findings and ML motivations above are calculated from the six Group030 standardised tables.